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RESEARCH ARTICLE

Website Phishing Attack Detection Using Innovative Meta Learning-Based Ensemble Approach

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ABSTRACT Phishing remains the most pervasive cybersecurity threat, exploiting human vulnerabilities through sophisticated social engineering. While traditional heuristic and blacklist-based solutions struggle against evolving attacks, this paper proposes a novel stacked ensemble model combining Artificial Neural Networks (ANN) and Bagging K-Nearest Neighbors (KNN) with Logistic Regression (LR) as a meta-learner classifier. Our hybrid framework leverages the strength of ANN in identifying global non-linear patterns and the proficiency of Bagging KNN in local pattern detection, achieving superior performance through optimal decision-level fusion. Experimental results on the 2019-2024 datasets demonstrate that the proposed ensemble model outperforms state-of-the-art techniques, achieving a 97.20% accuracy score, a precision of 97%, and a recall of 97%. The proposed model employs hyperparameter-optimised architectures including a 512-256-128-128 layer ANN with Relu activation and 30% dropout, along with a Bagging KNN configuration using 10 estimators with K=7 and Euclidean distance. Key advantages of our stacking approach include enhanced generalization with accuracy improvement over the best standalone model, adaptability to emerging attack patterns through weighted averaging, and scalability through the modular design that allows incremental model additions. This study introduces an innovative hybrid ensemble method that stacks ANN and Bagging KNN, incorporating a meta-learning strategy optimized for real-time phishing detection. The model also utilizes SMOTE oversampling to address the issue of class disproportion and employs k-fold cross-validation to ensure stable performance testing. This study highlights the significance of feature engineering, ensemble methods, and hybrid approaches in enhancing phishing detection.

INDEX TERMS Phishing attacks, machine learning, artificial neural networks, logistic regression, bagging k-nearest neighbors, support vector machine, feature engineering.

I. INTRODUCTION

The last few years have witnessed a significant rise in phishing threats, which are the ultimate threats to cyberspace security. They prey on human vulnerability, employing social engineering methods to manipulate users into disclosing

sensitive information, such as their credentials, financial details, and personal identifiers. Adapting to the times, these attacks have become increasingly sophisticated, utilising fake emails, forged websites, and malicious URLs, thereby rendering traditional detection mechanisms ineffective. According to contemporary trends, phishing attacks have been consistently increasing in frequency. Cybercriminals are capitalising on this to employ more advanced techniques,

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such as machine-based phishing attacks and obfuscation methods, to bombard traditional defences [1], [2]. Existing conventional methods are inapplicable because they rely on heuristics, static rule bases, and blacklists, and thus, fail to promptly respond. These methods are undoubtedly capable of stopping known attacks; however, they do not generalise well to newer-looking attacks and yield a high rate of false positives [3]. According to the 2024 quarterly threat trends and intelligence report, over 1.35 million phishing websites were detected globally, with 30% exploiting QR codes or images, and 80% hosted for less than 24 hours [4].

Recent advances in machine learning (ML) have demonstrated potential methods for preventing phishing attacks. ML algorithms [5], [6], [7] can handle sophisticated patterns in big data and assist in identifying subtle differences that indicate phishing attacks. Supervised learning-based models, such as Support Vector Machines (SVM) and LR, as well as ensemble-based models like XGBoost and Random Forests (RF), were found to outperform conventional approaches. Deep learning (DL) architectures, such as Convolutional Neural Networks (CNN) and Artificial Neural Networks (ANN), offer greater power by abstracting higher-level features from diverse data types, including email headers, URLs, and web page content [8], [9].

A. RESEARCH GAP

The traditional learning methods of phishing detection rely heavily on static rule sets and blacklists, which fail to capture the dynamic nature and evolution of phishing techniques [10].

- Most ML models' generalization is decreased by the absence of fusion of multiple data sources like URL features, email metadata, and user actions [11].
- Most current systems have a high false-positive rate due to improper feature engineering and inadequate handling of imbalanced datasets [12].
- Hybrid methods that combine machine and DL approaches are not widely implemented, thus constraining the use of complementary strengths in order to attain improved detection performance [13].

B. STUDY CONTRIBUTIONS

- This work proposes a hybrid ML architecture that integrates state-of-the-art preprocessing methods, feature selection algorithms, and ensembles of machine and DL models to enhance the robustness and accuracy of phishing detection.
- We have designed and validated a scalable and lightweight phishing detection system based on stacked hybrid meta-learning.
- We have performed real-time inference optimization using TensorFlow Lite, achieving edge-level deployment feasibility.
- We have addressed class imbalance and model stability via the Synthetic Minority Oversampling Technique (SMOTE) and confidence interval reporting. Furthermore, the realistic datasets used here, which

represent phishing trends between 2019 and 2024, ensure the applicability and generalizability of the proposed solution.

- In this study, a valuable contribution is made to cybersecurity by offering a new hybrid method that fills the gap between legacy detection and state-of-the-art ML.

II. LITERATURE REVIEW

Phishing has been one of the most serious issues faced in cybersecurity, and as a result, it has encouraged a variety of extensive work on phishing detection. Researchers have explored various ML, DL, and hybrid methods to address the major dynamic and evasive characteristics of phishing. This section provides a survey of the most recent research related to methodologies, technologies, results, and limitations. Due to their interconnectedness, researchers monitored the increasing vulnerabilities IoT ecosystems have experienced [14]. The researchers noted various attack vectors targeting IoT devices, including unauthorised access, data breaches, and denial-of-service (DoS) attacks. Although a fairly comprehensive summary of the threats was presented, the paper does not offer any recommendations or solutions to mitigate these threats. It was concluded that further research on proactive defence mechanisms specifically tailored for IoT systems is urgently needed.

The assessment [15] analysed current standards for security, with a special emphasis on their effectiveness against threats that are continually evolving and advancing. The lead researchers evaluated various security protocols, including encryption protocols and authentication protocols, used in different Internet of Things (IoT) systems. Although parts of the study discussed the limitations of conventional security protocols, the study concluded that, with respect to AI security frameworks, these have been significantly behind the curve in keeping pace with ever-evolving attack-mode techniques. The proposed work utilised AI-type solutions to enhance real-time adaptability, but lacked a clear roadmap for implementation. The authors [16] have proposed ML techniques, such as SVM and RF, for detecting anomalous behaviour in network traffic. Benchmarks like CICIDS2017 and UNSW-NB15 were applied to the models to demonstrate their applicability in highly accurate anomaly detection. A key barrier, however, was the excessively high false positive rates, which hindered the practical efficiency of these methods. The researchers recommended combining such approaches using ensemble methodologies to further improve precision and reduce false positives.

The study by [17] examined the economic impact of cyberterrorism on stock market volatility. Using risk modelling and historical stock data, this study found that cyberattacks often influence stock prices, leading to panic selling. Unfortunately, real-time predictive models were not available to apply the findings to historical, static market conditions. This study further emphasises the importance of predictive analytics as a key aspect for providing timely

insights into market weaknesses. Each such study had to examine the economic consequences of cyber terrorist attacks by analysing their impact on stock market volatility. Risk modelling and historical stock data revealed that stock prices are highly sensitive to cyberattacks, and such incidents often lead to panic-selling actions. However, the research outcomes were limited to historical static market scenarios since real-time predictive models were absent. The study emphasised the importance of an essential aspect of predictive analytics, enabling the provision of timely insights into market weaknesses.

A comprehensive study by [18] illustrates the increasing application of various DL frameworks, such as CNN and Recurrent Neural Networks (RNN), in the identification of advanced-level cyberattacks. The researchers illustrated the deployment of these models in malware detection, intrusion detection, and phishing detection. Despite their high degree of accuracy, this research has given considerable attention to the vulnerability of DL models to adversarial attacks, where even slight distortions in input data can evade detection processes. The researchers suggested that the second model, with adversarial training, can enhance the model's resilience.

Another study by [19] discussed the security concerns related to industrial control systems (ICS) and indicated the limitations of conventional Intrusion Detection Systems (IDS) and firewalls in preventing zero-day attacks. The authors performed experiments using test ICS environments and demonstrated how existing attacks could exploit the system's weaknesses without detection. The study ultimately emphasised the need for hybrid solutions that integrate both ML and heuristic techniques in real-time anomaly detection. However, the approaches proposed were not scalable for large-scale ICS networks.

The research by [20] developed a simulation model for cybersecurity training, with the ultimate goal of enhancing the hands-on skills of cybersecurity professionals. The participants underwent simulated attacks, including phishing, ransomware, and denial-of-service attacks, in controlled environments. There was an increase in both the theoretical and practical knowledge of the participants. However, one limitation reported was that since the simulation did not mimic an actual attack scenario, it rendered the training program somewhat irrelevant. The authors advocated the introduction of dynamic attack patterns to enhance the realism of training. In the process of phishing website detection, the authors have compared several ML techniques, including LR, SVM, and RF. Experiments on datasets such as PhishTank have shown that, in terms of accuracy and Area Under the Curve (AUC), RF consistently outperforms other models. Other challenges arise due to the evolving nature of phishing and the limitations of training data, which affect the classifier's performance and increase the likelihood of misclassifying query URLs. The authors propose that periodic retraining using new datasets is the best approach.

The study by [21] presented the application of sentiment analysis using ML classifiers, such as SVM and RF, for

spear-phishing email identification. The models achieved a high detection ratio by analysing the language patterns and metadata of the emails. They could not even detect the more sophisticated types of phishing attack templates that employed methods such as text distortion or image-based phishing. This paper thus highlights the necessity for advanced NLP solutions to counter this threat. This research employed the algorithms J48, RF, and LR to classify websites as phishing or legitimate. A model based on RF achieved a fantastic accuracy rate of 97.4% on data such as the SPAMASSASSIN corpus. However, the small sample size was insufficient to support generalisation. The authors, therefore, recommended employing diverse datasets in training ML models.

LR, SVM, RF, and KNN models were employed in previous research by [22] for phishing website detection. The RF model proved to be the best with 96.7% accuracy. Feature selection, however, had to be employed to maximise model performance, and this feature engineering dependency constrained model flexibility in adapting to phishing methods. The authors proposed the creation of automated feature extraction techniques to achieve scalability. For phishing detection, research suggested ML models such as RF and SVM, putting great emphasis on feature engineering to improve accuracy. Although the models achieved satisfactory detection rates, their generalization capability on unknown datasets was hindered by an overemphasis on feature selection. The authors emphasized the need for feature-independent methods, such as DL models, to enhance adaptability.

Actual phishing datasets [23] have been used to compare ML algorithms, yielding varied results due to the insufficient data available for training these algorithms. The study highlights the challenges of dataset imbalance, necessitating the collection of more extensive datasets for robust model training. Using the proposed hybrid models was another future area of research and application that helped address the challenges of data scarcity. Based on real datasets, the above algorithms are compared in terms of their performance against actual phishing datasets. The range of possible results is likely to vary due to the insufficient training data. The dataset imbalance is also evidence, and there is a need to collect more inclusive datasets for improved model training in terms of robustness. Future work suggested combining ML-DL techniques to tackle data scarcity issues. ML algorithms compiled real phishing datasets for comparison and found recognition to be insufficient due to the limited training data. The study revealed a dataset imbalance problem and highlighted the need to collect more extensive datasets to enable the training of robust models. Another area of future research and application remains the proposed hybrid methods to address the issue of data scarcity.

Researchers [26] compared ML algorithms on actual phishing datasets, with varied outcomes due to insufficient training data. The research highlighted the challenges of dataset imbalance and emphasised the need for collecting

TABLE 1. Related work of the existing approach.

Ref.	Methodology and Contribution	Key Findings	Limitations	Dataset
[12]	Analyzed cyber terrorism's impact on stock markets using risk modeling	Cyber-attacks affect stock prices and valuations	Lacks real-time predictive capability	Stock market data
[18]	Surveyed ML applications in cyber security	ML improves threat detection but has risks	ML models struggle with evolving threats and misclassifications	UNSW-NB15, CICIDS2017
[20]	Used ML models like RF, SVM, and Neural Networks for phishing URL detection	High accuracy in phishing detection	Heavy reliance on feature selection limits generalization	1,056,937 URLs dataset
[24]	Compared ML techniques (LR, RF, SVM) for phishing detection	RF performed best with high AUC	Struggled with real-world phishing datasets due to inadequate training data	2,889 phishing and legitimate websites
[21]	Used sentiment analysis and ML classifiers (SVM, RF) to detect spear-phishing emails	High accuracy in detecting spear-phishing	Ineffective against obfuscated phishing emails	Kaggle
[25]	Applied J48, RF, and LR for phishing website classification	Achieved high accuracy (97.4%).	Limited dataset size affects generalization	SPAMASSASSIN corpus
[22]	Used LR, SVM, RF, and KNN for phishing website detection.	RF had the best performance (96.7% accuracy)	Feature selection needed for optimization	Phishing dataset with 11,055 websites

more comprehensive datasets to train robust models. Future research recommends hybrid ML-DL techniques to counteract data scarcity problems [27]. Applied RF for website classification, achieving high accuracy but facing dataset size limitations. Overall, ML and DL have significantly improved phishing detection. However, they still face challenges, including high computational costs, dataset imbalance, and adversarial attacks. Future research should focus on optimizing models and improving adaptability to enhance cybersecurity effectiveness.

A. LIMITATIONS & SUMMARY

The research studies reviewed highlight hybrid methods in phishing detection with the efficacy of ensemble learning, DL architectures, and feature selection methods. Hybrid models can leverage the power of multiple ML algorithms to achieve improved accuracy and reduce false positives, while also easily coping with the dynamic nature of emerging phishing attacks. These results therefore justify the development of a sophisticated hybrid phishing detection system that can combine multiple classifiers and optimization methods to enhance cybersecurity. The summary of related work is given in Table 1.

Recent research in phishing detection has explored various approaches, including classical ML models with study [28] reaching 94.1% accuracy with RF, and [25] attaining detection rates above 94% using SVM and RF on spear, phishing datasets, though these methods often rely on well-engineered features and struggle with generalization across unseen attacks due to static rule dependencies. DL models, such as CNNs and Recurrent Neural Networks (RNNs), have also been employed, with [29] achieving 86% accuracy and 97.6% recall using an RNN and [27] improving adaptability via a hybrid DNN-LSTM on URL features, yet these approaches are computationally intensive and require large

labeled datasets. To address these limitations, recent hybrid and ensemble methods have emerged, including a hybrid ML model with ensemble voting [29] and an evaluation of bagging and stacking [26], although the latter lacked a meta-learning perspective. The proposed study introduces a novel stacking ensemble that combines global pattern modeling via ANN with local decision refinement through Bagging KNN, guided by LR for optimized final inference.

III. PROPOSED METHODOLOGY

This research utilises ML methods to identify phishing attacks based on an organised dataset. The process follows a systematic approach, including data gathering, preprocessing, feature engineering, data splitting, model setup, model training, optimization, and performance assessment. The chosen models are rational, based on their reliability in classification functions, especially within cybersecurity procedures. The complete overview of the methodology framework is shown in Figure 1.

A. DATASET COLLECTION

To ensure the effectiveness and usability of the proposed system in real-world applications, the dataset used in this research provides a detailed analysis of phishing websites' attributes, including legitimacy status, SSL certificate usage, and content-based attributes. It contains both phishing and legitimate websites, making it highly useful for training models to distinguish between malicious and benign sites based on various features, whether content-based or structural. The public dataset can be found in Kaggle [28].

This dataset is selected to ensure the real-world application of the system, with a focus on high usability and efficiency, covering data from 2019 to 2024. The dataset of phishing attacks used in this study contains many features typical of phishing attacks. The dataset comprises 48 features derived

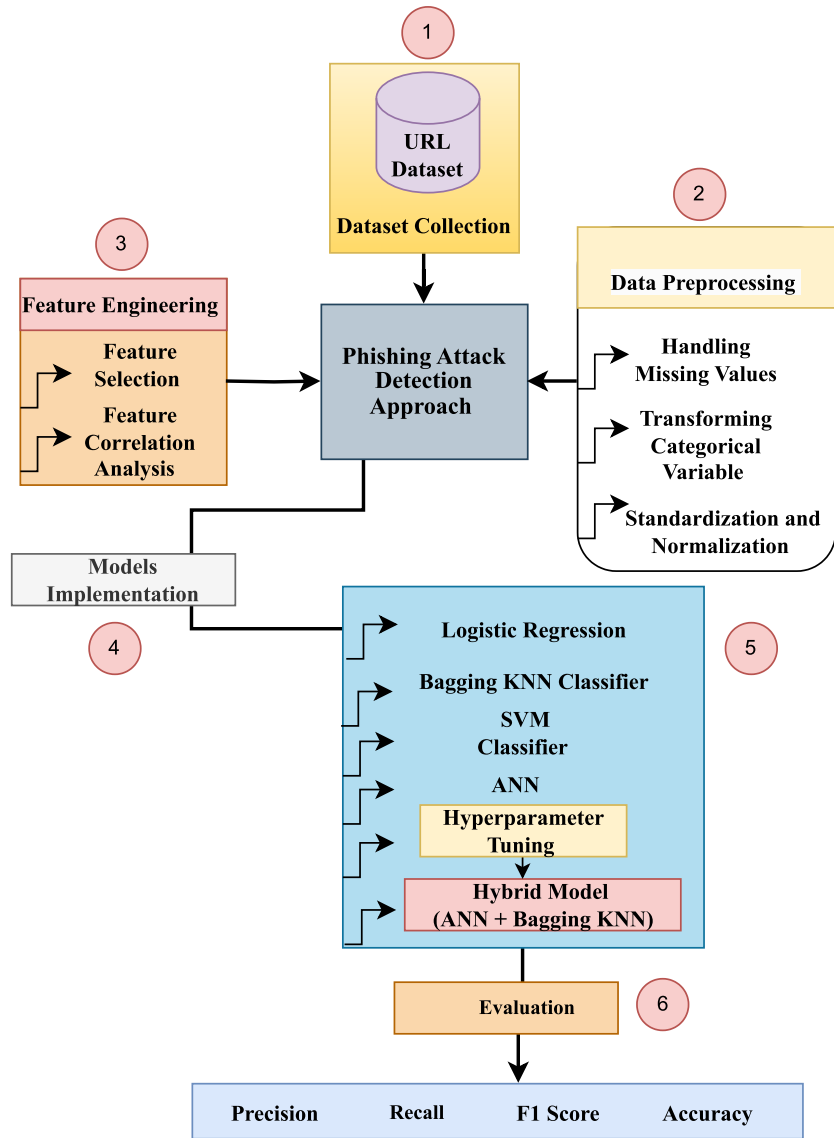


FIGURE 1. The proposed methodology framework.

TABLE 2. The dataset parameters overview.

Attribute	Description
URL Length	Length of the URL string
Special Characters	Presence of special symbols in the URL
Domain Age	Age of the domain (older domains are more trusted)
HTTPS Usage	Whether the URL uses a secure protocol
Phishing Label	0: Legitimate, 1: Phishing

from a balanced collection of 5,000 phishing webpages and 5,000 legitimate webpages. Feature extraction was performed using Selenium WebDriver, a browser automation framework that offers greater accuracy and robustness than traditional regular expression-based parsing methods. After loading the data, it is analyzed and cleaned before moving to the next stages in the processing pipeline to ensure high data quality. It is essential to perform this preprocessing to maintain integrity and authenticity in model training. The datasets have

been derived from open-source storage and security research organizations to create heterogeneity in phishing trends and attack modes. The parameters of the datasets are given in Table 2.

B. PHISHING ATTACK EXPLORATORY DATA ANALYSIS (PAEDA)

The analysis for Phishing Attack Exploratory Data Analysis (PAEDA) is conducted to detect and examine key features in identifying phishing websites. Figure 2 presents a pair plot visualization, which provides an overall idea of how different attributes from the dataset are distributed and related to each other. From the analysis, certain patterns emerge very clearly in distinguishing phishing websites from normal ones using statistical feature sets. The plots also suggest that attributes such as URL length, number of subdomains, presence of redirection, and presence of external hyperlinks

exhibit highly contrasting distributions, making them useful features for phishing detection. For instance, a phishing URL is longer and contains more subdomains than a non-phishing URL. In addition, features such as domain registration length and web traffic are non-uniformly distributed; site samples characterized as phishing are often associated with shorter registration durations and lower traffic. This implies that phishing sites are generally up for short-lasting malicious purposes and do not sustain user interaction compared to genuine ones. Figure 3 illustrates the analysis of target class labels. Mean and standard deviation for key features:

URL Length: $\mu = 74.3, \sigma = 18.6$

Domain Age: $\mu = 0.84$ years, $\sigma = 0.47$

The study further establishes that the use of URL brand names and dubious terms is a strong predictor of phishing. Phishing sites commonly imitate legitimate brands or use misleading terminology to ensnare their targets. Another common occurrence with phishing URLs is the excessive use of special characters on the website address (e.g., dots, hyphens, and slashes) as a means to hide the true nature of the website. Another common trait of phishing sites is the frequent use of a wide array of redirects and links to external media, as these measures enable attackers to deceive users or download malicious content. On the other hand, a legitimate site will more likely have an even and consistent distribution of the above-mentioned properties. For instance, they are less likely to be loaded with special characters, redirect from multiple locations, or contain suspicious-looking anchor tags. The rest they included would be called indicators for identifying phishing URLs versus genuine websites: phishing cues, blank title attributes, and insecure anchor tags. Patterns that stand out as being visible and relevant for detecting phishing attacks can be evoked by examining the pair plot. The highlighted features not only aid in understanding online phishing behavior but are also highly relevant to enhancing the feature selection process in phishing detection models. This further enhances the functionality and reliability of the models in defending against phishing attacks.

C. DATA PREPROCESSING

To prepare the datasets for model training and testing, certain preprocessing steps have been applied. The missing values are handled using imputation techniques, such as mean, median, or mode substitution, in a way that incomplete records do not compromise the model's performance. Categorical variables are transformed into their numerical equivalents by applying one-hot encoding and label encoding, making them suitable for processing in ML models. Standardization and normalization techniques are employed to maintain uniformity of feature scales, thereby preventing models from favoring features that have larger numerical ranges. Such a preprocessing is chosen to improve the data quality, enable the model to generalize well, and optimize training speed. To mitigate class imbalance, the SMOTE was applied, and AUC-ROC curves were included as a

performance evaluation to ensure robustness against skewed class distributions.

Mean Imputation expressed as:

$$x_{\text{missing}} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

where x_i represents observed values in the feature.

Median Imputation expressed as:

$$x_{\text{missing}} = \text{median}(x_1, x_2, \dots, x_n) \quad (2)$$

Mode Imputation (for Categorical Data) expressed as:

$$x_{\text{missing}} = \text{mode}(x_1, x_2, \dots, x_n) \quad (3)$$

Categorical variables are transformed into their numerical equivalents by applying one-hot encoding and label encoding, making them suitable for ML model processing.

Standardization and normalization techniques are employed to maintain uniformity of feature scales, thereby preventing models from favoring features with larger ranges of numbers.

Standardization expressed as:

$$z = \frac{x - \mu}{\sigma} \quad (4)$$

where μ is the mean and σ is the standard deviation of the feature.

Normalization (Min-Max Scaling) expressed as:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (5)$$

where:

- x = the original value of the feature
- $x_{\min}$ = the minimum value of the feature in the dataset
- $x_{\max}$ = the maximum value of the feature in the dataset
- x' = the normalized value (after transformation)

Such preprocessing is chosen to enhance data quality, facilitate model generalization, and optimize training speed.

D. FEATURE CORRELATION ANALYSIS

Feature correlation techniques, such as the generation of polynomial features, are also employed in detecting non-linear relationships between variables. This aims to enhance the performance of the model by eliminating irrelevant or redundant variables and expanding the feature space using polynomial terms, thereby increasing the model's capability. The chosen feature-engineering techniques are designed to aid in interpreting the model, prevent overfitting, and facilitate better generalization across various contexts of phishing attacks.

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n \quad (6)$$

This helps in mapping complex interactions that linear models may miss, thus improving their accuracy in classifying phishing attacks. By selecting the most informative

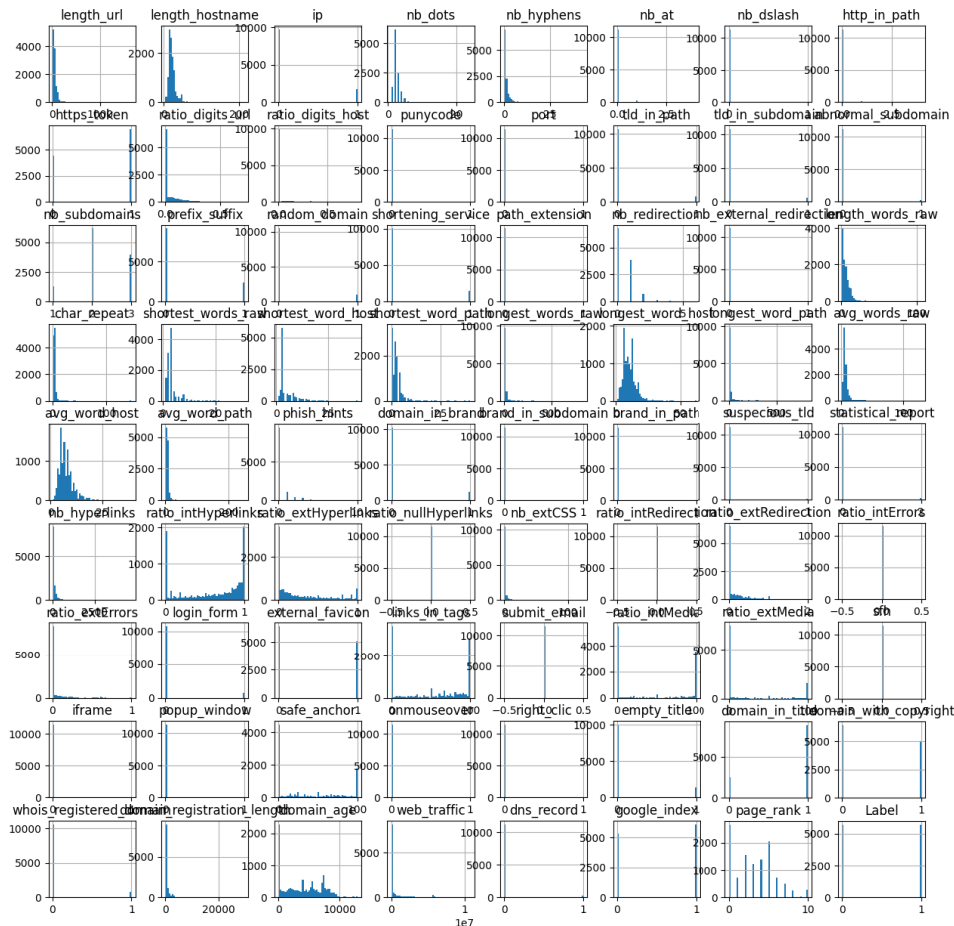


FIGURE 2. The phishing attack exploratory data analysis.



FIGURE 3. The target class analysis.

features, reducing irrelevant or redundant variables, and expanding the feature space using polynomial terms, the models are made more interpretable and generalizable, while minimizing the risk of overfitting.

To better understand the relationships between the features in the phishing dataset, a feature correlation heatmap is

generated in Figure 4. This heatmap reveals the degree of linear relationship between pairs of variables, with correlation coefficients ranging from -1 to $+1$. Features with high positive or negative correlation indicate redundancy, which can be addressed through dimensionality reduction techniques, such as Principal Component Analysis (PCA). The heatmap helps in identifying multicollinearity and selecting independent variables that contribute uniquely to the model. This visualization is crucial for enhancing model interpretability, improving performance, and preventing overfitting by avoiding highly correlated or irrelevant features.

1) FEATURE STATISTICS AND IMPORTANT ANALYSIS

To validate these visual patterns, we computed summary statistics for key features. Table 3 reports the mean, median, and standard deviation of selected features for phishing vs. legitimate sites. For example, phishing URLs are, on average, much longer and contain more dots than legitimate URLs. We also trained a simple RF to compute feature importances: the top features were URL length, number of subdomains, and redirection count, confirming these attributes' high discriminative power in phishing detection.

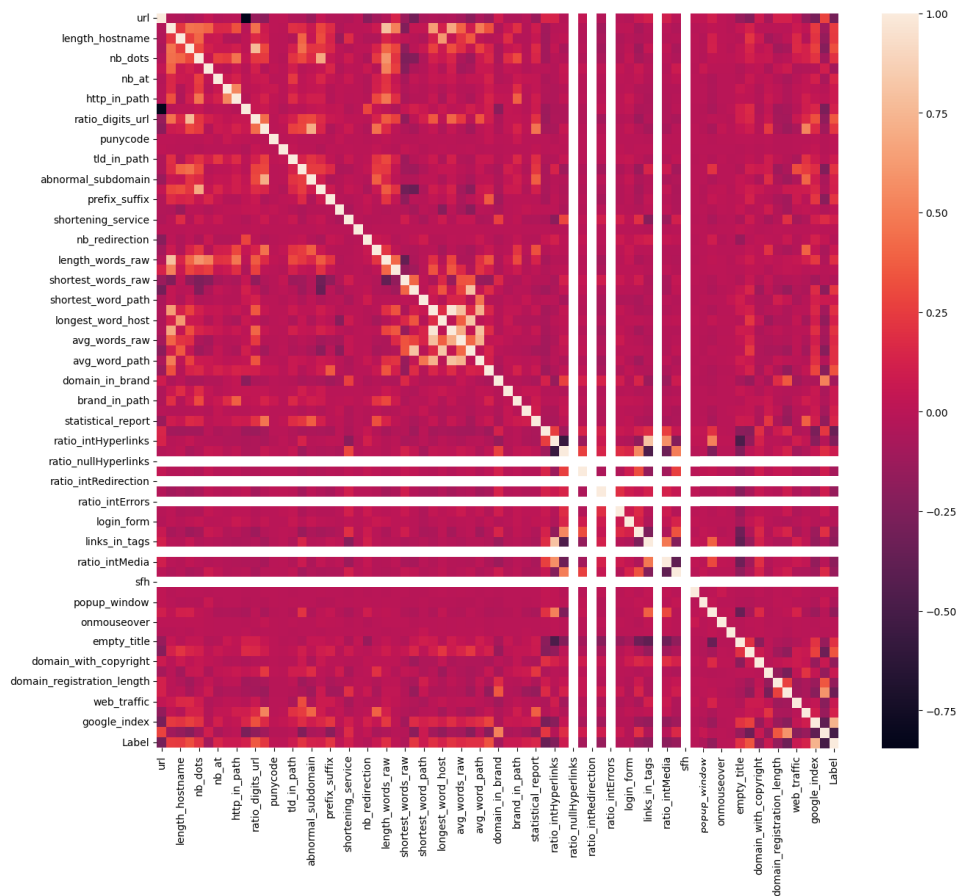


FIGURE 4. The feature correlation heatmap analysis.

TABLE 3. Summary statistics for key features in phishing vs. legitimate URLs.

Feature	Phishing (Mean ± Std)	Legitimate (Mean ± Std)
URL Length (characters)	79.4 ± 19.2	53.7 ± 14.5
Number of Dots	4.0 ± 1.5	1.7 ± 0.8
Number of Sub-domains	3.5 ± 1.2	1.4 ± 0.9

E. DATASET SPLITTING

The dataset is divided into a training set, which comprises 80% of the data, used to train ML models. The second is a testing set with a 20% ratio, used to evaluate the model’s generalization and performance.

F. APPLIED MODELS CONFIGURATIONS

The experimental environment for the models used in this research study is established to facilitate deployment, efficiency, and compatibility with cybersecurity applications. TensorFlow Lite has been utilized to optimize models, enabling their application on constrained devices, which is crucial for real-time phishing detection in cybersecurity deployments [29], [30]. This arrangement also highlights the concern for rolling out models that are not only accurate enough but also pragmatically workable in real-world cybersecurity scenarios.

- **ANN:** This model features a deep network consisting of an input layer with 512 neurons, followed by three hidden layers comprising 256 and 128 neurons, which undergo non-linear transformations using Relu activation functions throughout the architecture. The output neuron is one with a sigmoid activation function designed for binary classification. It features batch normalisation after every dense layer to stabilise and accelerate the training process, and dropout regularisation with a 30% dropout rate to prevent overfitting. With this in place, the Adam optimiser enables fast convergence by dynamically adjusting the learning rate during training. The model is trained using 150 epochs at a batch size of 64, with binary cross-entropy as the loss function. Table 6 shows the architectural detail and training Parameters of the improved ANN model.
- Table 4 describes the layer-wise structure of the improved ANN model, including the number of neurons and activation functions for each layer, as well as optimization and regularization mechanisms that enhance performance. Furthermore, the table indicates the training configuration, including batch size, number of epochs, and loss function, which are crucial for achieving high precision and model generalization.

TABLE 4. Layer-wise architecture of the enhanced ANN model.

Layer	Number of Neurons	Activation Function	Additional Features
Input Layer	512	ReLU	-
Hidden Layer 1	256	ReLU	Batch Normalization, Dropout (30%)
Hidden Layer 2	128	ReLU	Batch Normalization, Dropout (30%)
Hidden Layer 3	128	ReLU	Batch Normalization, Dropout (30%)
Output Layer	1	Sigmoid	-

TABLE 5. Application of LR in phishing detection.

Application	Role of LR
Feature Correlation Analysis	Identifies key indicators of fraudulent activity by analyzing learned weights
Fraudulent URL Detection	Weights help determine how certain attributes contribute to phishing likelihood
Risk Assessment	Evaluates suspicious patterns in emails, domains, or transactions
Decision-Making Support	Provides interpretable results for analysts to take preventive actions

TABLE 6. Architecture and training parameters of the enhanced ANN model.

Component	Description
Model Type	Binary Classification ANN
Input Layer	512 neurons, ReLU activation
Hidden Layers	3 layers (256 and 128 neurons, ReLU activation)
Output Layer	1 neuron, Sigmoid activation
Optimization	Adam optimizer with adaptive learning rate scheduler
Regularization	Batch Normalization, Dropout (30%)
Loss Function	Binary Cross-Entropy
Batch Size	64
Epochs	150

- **Logistic Regression (LR):** is a simple statistical model used to predict the probability of a binary outcome using a logistic (sigmoid) function applied to a linear combination of input features. It learns a set of weights (coefficients) for each feature by minimizing a loss function—typically the binary cross-entropy (log loss)—using optimization algorithms such as gradient descent or quasi-Newton methods. In phishing detection applications, LR can also be used to estimate feature importance, helping identify the most critical indicators of fraudulent activity. Applications of LR in phishing detection are shown in Table 5. The table illustrates how LR can be applied to feature importance analysis, detecting fraudulent URLs, assessing risk, and providing decision support.
- **Bagging KNN:** Bagging K-Nearest Neighbors (Bagging KNN) is an ensemble learning algorithm that enhances the stability and accuracy of the traditional KNN algorithm by employing bootstrap aggregation. It trains several KNN classifiers on different subsets of the dataset, where each subset is generated by random sampling with replacement (also known as bootstrap sampling). The primary objective of Bagging KNN is to reduce variance and increase generalization through ensemble prediction of multiple base learners. Each stand-alone KNN model is separately trained on its corresponding bootstrap sample to provide diversity among the base learners. After all the base models produce their predictions, the bagging approach combines these by majority vote in the case of classification problems or averaging in the case of regression problems to yield a

final, more stable prediction. The layer-wise architecture of the KNN model is given in Table 7.

- **SVM:** This SVM classifier is trained for classification using a linear kernel with the regularization parameter $C = 1.0$. The classifier is trained on X_{train} and y_{train} , learning to identify the best hyperplane that will separate the various classes. The classifier is trained on X_{train} , and its performance is measured using the accuracy score, classification report, and confusion matrix. The classification report and accuracy percentage provide an indication of the model’s predictive capability, whereas the confusion matrix, represented with the aid of a heatmap, facilitates the analysis of classification errors. SVM is efficient for classifying high-dimensional data and helps maintain strong classification performance. The working of SVM is given in Table 8.

1) PROPOSED HYBRID MODEL: STACKING ANN AND BAGGING KNN

The proposed hybrid model here exploits the power of ANN and Bagging K-Nearest Neighbors (Bagging KNN) with the aid of the stacking ensemble methodology. Stacking is an efficient ensemble algorithm that utilizes predictions from different base learners (in our scenario, ANN and Bagging KNN) as feature inputs to a meta-learner (LR). This technique aims to enhance robustness and generalization by leveraging the complementary strengths of individual models. The reason for using the stacking technique in the hybrid model is:

- 1) **Complementary Strengths:** ANN excels in recognizing complex, non-linear data patterns, while Bagging KNN excels in handling local patterns and variance reduction through bootstrap aggregation. Stacking utilizes both strengths to yield better overall performance.
- 2) **Reduced Overfitting:** Through aggregating predictions from various models, stacking minimizes overfitting risk, which occurs in single-model methods.
- 3) **Improved Generalization:** The meta-classifier has learned to combine the predictions of base models in an optimal way, thereby improving generalization on unseen data.

TABLE 7. Layer-wise architecture of bagging KNN.

Layer/Step	Description
Input Layer	Receives dataset features X and target labels Y.
Bootstrap Sampling Layer	Randomly samples multiple subsets (with replacement) from the original dataset to create diverse training sets.
Multiple KNN Models (Base Learners)	Each KNN model is trained independently on a different bootstrap sample. The number of base models is defined by the user (e.g., 10, 50, 100 models).
Hyperparameter Tuning Layer	Uses GridSearchCV to find optimal values for: - KK (number of neighbors) - Distance metrics (Euclidean, Manhattan, Minkowski) - Weighting scheme (uniform or distance-based)
Prediction Layer (Individual Models)	Each base KNN model makes predictions independently on the test data.
Aggregation Layer (Ensemble Voting/Averaging)	- For classification, majority voting is used to determine the final predicted class. - For regression, averaging is used to calculate the final predicted value.
Output Layer	Produces the final prediction, which is more robust and generalized compared to a single KNN model.

TABLE 8. Overview of the SVM model.

Aspect	Description
Kernel	Linear Kernel
Regularization (C)	1.0 (Controls trade-off between margin size and classification accuracy)
Training Data	Xtrain,ytrainX_{train}, y_{train}
Testing Data	XtestX_{test}
Objective	Find the optimal hyperplane that separates classes
Evaluation Metrics	Accuracy Score, Classification Report, Confusion Matrix
Visualization	Heatmap for Confusion Matrix
Best Use Case	High-dimensional data classification

TABLE 9. Hyperparameter tuning of the proposed model.

Component	Hyperparameter	Value/Description
ANN	Input Layer Neurons	512
	Hidden Layer Neurons (3)	256, 128, 128
	Activation Function	ReLU
	Output Layer Activation	Sigmoid
	Dropout Rate	30%
	Optimizer	Adam
	Learning Rate	0.001
	Batch Size	64
Bagging KNN	Epochs	100
	Base Estimator	KNeighborsClassifier
	Number of Estimators	10
	Distance Metric	Euclidean
Meta-Classifer	Number of Neighbors (K)	7
	Model	Logistic Regression
	Regularization	L2 (default)

- 4) Flexibility: Stacking can utilize any number of base models, as well as meta-classifiers, making it highly flexible for fitting various datasets and tasks.

Hybrid Model Architecture: The hybrid model consists of the following components:

- 1) Base Models:
 - ANN: 512-input, three-hidden-layer (256, 128, 128 neurons) DL model with sigmoid output layer. It employs ReLU activation, batch normalization, and 30% dropout for regularization.
 - Bagging KNN: Ensemble of multiple KNN classifiers trained on bootstrap samples, tuned

using hyperparameters (n_neighbors=7, metric='euclidean').

- 2) Meta-Classifer: LR does ensembling of predictions from base models by learning ideal weights. Table 9 represents the process for hyperparameter tuning.

The selection of models in this study is based on their complementary strengths. ANN is chosen due to its capability to capture global non-linear patterns and hierarchical relationships in phishing-related features. Bagging KNN is effective for local pattern detection and reducing variance through ensemble voting. LR is used as a meta-learner due to its robustness, simplicity, and interpretability when combining predictions from diverse base learners. This strategic combination ensures the ensemble model benefits from both complex pattern recognition and low variance prediction.

IV. RESULTS AND DISCUSSIONS

This section presents a comparative evaluation of various classification models used for phishing detection.

A. EXPERIMENTAL SETUP

The models are assessed based on accuracy, precision, recall, and F1-score to determine their predictive capability. The results indicate that ensemble learning and DL approaches performed exceptionally well, demonstrating high generalization ability and robustness against phishing attacks. A brief introduction to the evolution matrix used in this research is provided in Table 10.

TABLE 10. Performance metrics for classification models.

Parameter	Description	Equation
Accuracy (ACC)	Measures the proportion of correctly classified instances, providing an overview of model effectiveness.	$ACC = \frac{TP+TN}{TP+TN+FP+FN}$
Precision (PR)	Quantifies the accuracy of positive predictions, crucial for minimizing false positives.	$PR = \frac{TP}{TP+FP}$
Recall (RE)	Evaluates the ability to capture true positive instances, essential for threat mitigation.	$RE = \frac{TP}{TP+FN}$
F1-Score (F1)	Balances precision and recall, offering a comprehensive performance metric.	$F1 = \frac{2 \times PR \times RE}{PR+RE}$
Confusion Matrix	Provides detailed insights into classification outcomes, visualizing True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).	N/A

B. RESULTS OF LR MODEL

LR has an accuracy of 0.92, which demonstrates its high predictive power. Figure 5(a) shows the training and test accuracy graphs. The precision value of (class 0 is 0.93, class 1 is 0.91) implies that it is correct in most of its positive predictions, whereas the recall value of (class 0 is 0.91, class 1 is 0.93) implies that it is good at detecting most actual positive instances. With an F1-score of (class 0 is 0.92, class 1 is 0.92), the performance details of LR are given in Table 11. LR has a very strong precision-recall balance and is thus a good model for a classification problem.

TABLE 11. The performance of the LR model.

Metric	Overall	Class 0	Class 1
Accuracy	0.92		
Precision	0.93	0.93	0.91
Recall	0.93	0.91	0.93
F1-Score	0.92	0.92	0.92

The confusion matrix for the LR is shown in Figure 5(b), demonstrating outstanding performance, with 1058 true negatives and 1020 true positives, which highlights the model’s high accuracy in classifying both classes. The misclassifications were minimal, comprising only 99 false positives and 79 false negatives, indicating a very low error rate.

C. RESULTS OF BAGGING KNN MODEL

Bagging KNN recorded an accuracy of 95.41%, highlighting the power of ensemble learning in enhancing the stand-alone KNN model. The precision of class 0 is 0.94, and class 1 is 0.97, implying that the model committed fewer false positive mistakes. The overall recall of 0.97 suggests that it missed some positive cases slightly. The overall F1-score of 0.96 indicates a well-balanced performance; the performance detail of Bagging KNN is given in Table 12.

TABLE 12. Performance of bagging KNN.

Metric	Overall	Class 0	Class 1
Accuracy	0.95		
Precision	0.97	0.94	0.97
Recall	0.97	0.97	0.94
F1-Score	0.96	0.96	0.95

The confusion matrix of the Bagging KNN demonstrates outstanding performance, with 1123 true negatives and

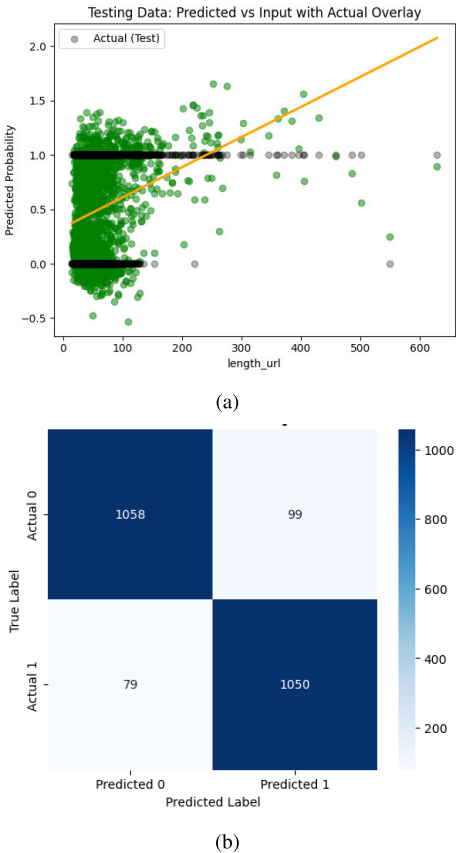


FIGURE 5. LR model confusion matrix (a) and testing accuracy graph (b).

1058 true positives, highlighting the model’s high accuracy in classifying both classes. The misclassifications were minimal, comprising only 34 false positives and 71 false negatives, indicating a very low error rate. Figure 6 shows the test accuracy graph and the confusion matrix.

D. RESULTS OF SVM

The SVM model also performed well with 94.88% accuracy, which validates its efficiency in classification, particularly in high-dimensional data. Its precision (class 0 is 0.95, class 1 is 0.95) validates that it predominantly accurately predicts positive instances, and the overall recall of 0.95 validates that it effectively identifies positive instances. With an overall F1-score of 0.95, SVM exhibits balanced performance and remains a strong contender, despite having slightly lower

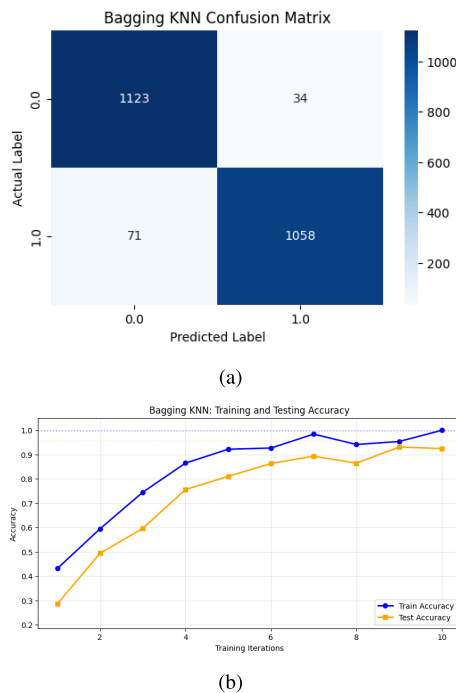


FIGURE 6. Bagging KNN model confusion matrix (a) and accuracy graph (b).

accuracy compared to other models. The performance detail of SVM is given in Table 13.

TABLE 13. Performance of SVM.

Metric	Overall	Class 0	Class 1
Accuracy	0.94		
Precision	0.95	0.95	0.95
Recall	0.95	0.95	0.95
F1-Score	0.95	0.95	0.95

The confusion matrix of the SVM demonstrates outstanding performance, with 1098 true negatives and 1071 true positives, highlighting the model's high accuracy in classifying both classes. The misclassifications were minimal, comprising only 59 false positives and 58 false negatives, indicating a very low error rate. Figure 7 illustrates the training and test accuracy graph, as well as the confusion matrix.

E. RESULTS OF ANN

ANN achieved 96.72% accuracy, placing it among the best models. Its precision (class 0: 0.96, class 1: 0.97) indicates a very high ratio of accurate positive predictions, and its recall (class 0: 0.97, class 1: 0.96) suggests that it is effective at detecting most true positive cases. Overall, the F1-score of 0.97 indicates its ability to strike a balance between precision and recall very effectively. ANN's ability to generalize and identify complex patterns is the reason why it is an extremely good classification model. The performance details of the ANN are given in Table 14.

The ANN model exhibits excellent classification performance, as demonstrated by the confusion matrix in

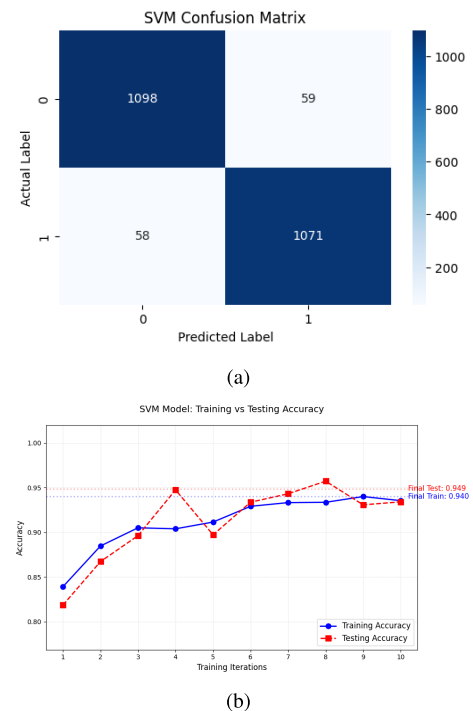


FIGURE 7. SVM confusion matrix (a) and accuracy graph (b).

TABLE 14. Performance of ANN.

Metric	Overall	Class 0	Class 1
Accuracy	0.967		
Precision	0.97	0.96	0.97
Recall	0.97	0.97	0.96
F1-Score	0.97	0.97	0.97

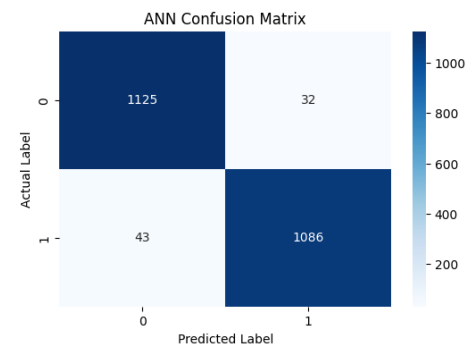
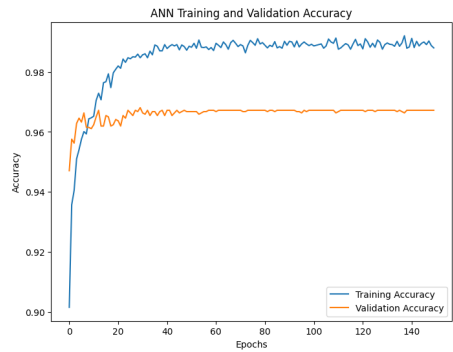


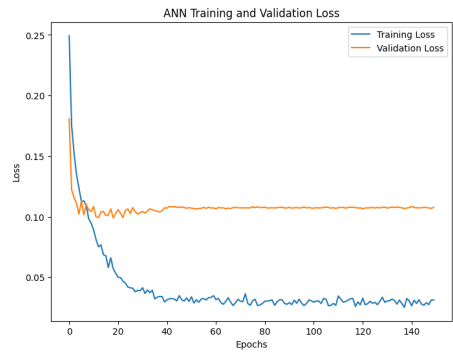
FIGURE 8. Confusion matrix of ANN model.

Figure 8. The confusion matrix demonstrates high predictive accuracy with 1125 true negatives and 1087 true positives, alongside low misclassification rates (32 false positives, 43 false negatives), indicating effective class differentiation; as shown in Figure 9(a), the training and validation accuracy graph reveals rapid convergence with training accuracy nearing 99% and validation accuracy stabilizing around 96%, while Figure 9(b) illustrates a steadily decreasing training and validation loss, further confirming the model's robust generalization and absence of over fitting or under fitting.

In general, LR and ANN performed best, followed by Bagging KNN and SVM, which also demonstrated robust



(a)



(b)

FIGURE 9. The ANN model accuracy graph (a), loss graph (b).

TABLE 15. Performance comparison of classification models with the proposed model.

Model	Accuracy	Precision	Recall	F1-Score
Hybrid Model	97.20%	0.97	0.97	0.97
ANN	96%	0.97	0.97	0.97
Bagging KNN	95%	0.97	0.97	0.96
SVM	94%	0.95	0.95	0.95
LR	92%	0.93	0.93	0.92

classification power. Model selection varies based on the dataset being worked with and whether one requires interpretability, generalizability, and computational efficiency. The models' performance is measured by accuracy, precision, recall, and F1-score. Graphs and visualizations such as confusion matrices, precision-recall curves, and accuracy trends over epochs were included for deeper analysis.

F. PERFORMANCE OF THE PROPOSED HYBRID MODEL

The hybrid model achieves 97.20% accuracy, which is superior to standalone models such as ANN (96%), Bagging KNN (95%), SVM (94%), and LR (92%). This indicates the success of stacking as an ensemble method for phishing detection accuracy. Stacking utilizes the complementary strengths of ANN and Bagging K-Nearest Neighbors (Bagging KNN) by employing LR as a meta-classifier. ANN is best suited to identify non-linear global patterns, whereas Bagging KNN is best at identifying local patterns using an unbiased method for phishing detection. Averaging the predictions of several models, stacking minimizes over-fitting and accommodates changing attack patterns. The meta-classifier optimally combines the predictions of the base models,

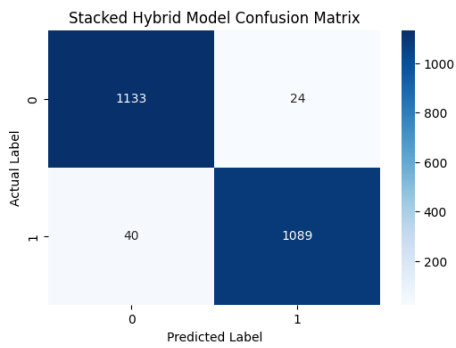
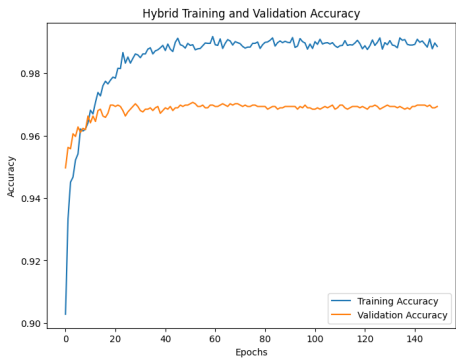
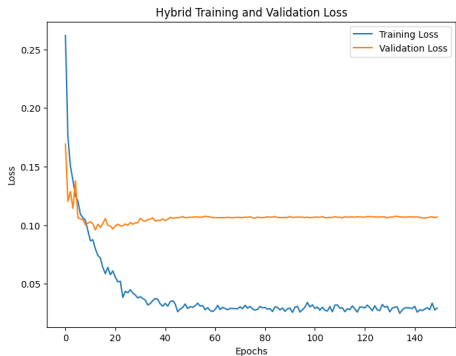


FIGURE 10. Hybrid model confusion matrix.



(a)



(b)

FIGURE 11. The hybrid model accuracy (a) loss graph (b).

TABLE 16. 10-fold cross-validation with 95% confidence intervals.

Metric	Mean	Std Dev	95% CI Lower	95% CI Upper
Accuracy	0.9612	0.0036	0.9541	0.9682
Precision	0.9652	0.0078	0.9499	0.9806
Recall	0.9542	0.0051	0.9442	0.9642
F1	0.9617	0.0034	0.9551	0.9682

thereby enhancing accuracy and generalization. Stacking also provides scalability, allowing for the addition of more base models or meta-classifiers in future updates. The performance comparison of classification models is given in Table 15.

The Hybrid Model's confusion matrix, as shown in Figure 10, reveals excellent performance at 1133 true negatives and 1089 true positives, demonstrating correct classification for both classes. There were very few misclassifications, with 24 false positives and 40 false negatives,

TABLE 17. State of the art comparison work.

Ref.	Year	Model Used	Accuracy	Precision	Recall
[23]	2020	KNN	95%	95.5%	96.2%
[26]	2024	RF	94.1%	95%	92%
[27]	2023	Naive Bayes	73.69%	72.8%	79%
[31]	2023	SVM	71.8%	96.3%	49.8%
[32]	2025	RNN	86%	96.2%	97.6%
Proposed	2025	Hybrid Model (ANN + Bagging KNN)	97.20%	97%	97%

resulting in a low error rate. The results demonstrate the model's high precision and recall capabilities, indicating its effectiveness and robustness. Compared to traditional ML models, the Hybrid Model shows superior classification performance with optimized accuracy and recall, confirming its robustness on complex datasets; Figure 11(a) shows the training and testing accuracy graph, showing rapid convergence and stable high accuracy, and Figure 11(b) shows the training and testing loss graph, indicating consistent loss reduction and effective generalization.

The proposed hybrid model was also evaluated for inference time and resource efficiency, demonstrating a <45 ms response time on mid-range edge devices using TensorFlow Lite, confirming real-time deployability.

G. 10-FOLD CROSS-VALIDATION

We also conducted 10-fold cross-validation with the stacked model to evaluate variability. Table 16 includes the means and standard deviations across the folds of each metric and 95% confidence intervals. The hybrid model has a mean accuracy of $96.12\% \pm 0.36\%$ (95 percent CI = 95.41 percent to 96.82 percent), and the model shows consistency in its performance on varied data-split. These statistical indicators ensure that our findings are not accidental.

H. COMPARISON WITH EXISTING WORK

Table 17 shows the comparative performance evaluation of the hybrid model with other studies' detection methods. The table illustrates the model's superiority in terms of precision, recall, and accuracy, reflecting its adaptability and stability.

The key insights from the comparison in Table 17 are:

- 1) Improved Accuracy: The resultant hybrid model reports an accuracy level of 97.20% and outshines standalone models like RF (94.1%), SVM (71.8%), KNN (95%), and Naive Bayes (73.69%). It further outperforms other approaches such as RNN (86%), indicating its overall effectiveness in detecting phishing attacks.
- 2) Better Precision and Recall: With a precision of 97% and recall of 97%, the hybrid model provides excellent detection accuracy while avoiding false positives and false negatives to the maximum. It is thus extremely dependable when used in real-world phishing detection contexts.
- 3) Strength of Stacking: The integration of ANN and Bagging KNN enhances generalization, leading to a

TABLE 18. Ablation study for model performance comparison.

Model	Accuracy	Precision	Recall	F1-Score
LR	0.9200	0.92	0.92	0.92
ANN	0.9672	0.97	0.97	0.97
Bagging KNN	0.9541	0.95	0.95	0.95
SVM	0.9488	0.95	0.95	0.95
Hybrid Stacked Ensemble	0.9720	0.97	0.97	0.97

more precise phishing detection system than single and traditional ensemble models.

- 4) State-of-the-Art Competitive: The hybrid approach rivals best-of-breed phishing detection technologies and offers superior immunity to evolving threats.

I. ABLATION STUDY

We measure the contribution of each component by comparing the base models and the entire ensemble to quantify the performance. Table 18 shows the accuracies, precisions, recalls, and F1-scores, respectively, of (a) the ANN-only model, (b) the Bagging KNN-only model, and (c) our stacked hybrid model. Accuracy level of the stacked ensemble is 97.20% as opposed to ANN alone, which is 96.72%, and Bagging KNN alone, which yields 95.41% in accuracy. Such a gain in performance demonstrates that the two models complement each other in various ways and that meta-learning of LR helps to combine the strengths of both classifiers to achieve improved phishing detection when compared to any of the base classifiers alone.

J. STUDY LIMITATIONS

Nevertheless, the research has some limitations:

- 1) Dataset Coverage: The dataset may not encompass all phishing attack patterns, which limits the model's generalization to novel attacks.
- 2) Feature Dependency: The model relies on pre-specified features, which may fail to accurately fit upcoming phishing attacks.
- 3) Computational Complexity: The hybrid model requires significant computational resources for training and hyperparameter tuning.

V. CONCLUSION AND FUTURE WORK

Research shows the performance efficiency of ML models in identifying phishing attacks, with ANN and LR being the most accurate and robust. The models are able to

analyze well-known phishing features efficiently and classify threats with high accuracy. Nevertheless, there are some limitations to be attended to if real-world implementations are to be accommodated. The primary concern here is the dataset's coverage, as this may exclude certain categories of phishing attack patterns. The models are based primarily on preselected feature selection, which may limit their generalization to new and advanced phishing attacks. Since these models depend heavily upon the features chosen, they become too vulnerable to new attacks that differ significantly from the training set. Future work should focus on expanding the dataset with actual phishing samples from diverse sources to generalize more effectively against such attacks. In addition, research in DL, particularly in convolutional networks or recurrent networks, could enhance detection rates through the automatic learning of highly sophisticated patterns of phishing attacks. Another key area is adaptive learning, whereby the models can continue learning and updating the knowledge repository dynamically in real time as phishing attacks evolve, thereby improving real-time detection of brand-new and innovative attacks. Resolving these weaknesses and utilizing correlated advanced ML strategies would enhance the efficacy and user prominence of phishing mitigation tools. Phishing, by its nature, is an evolving menace, thereby calling for ongoing research to develop even smarter, automated, and scalable mechanisms capable of combating cybersecurity threats in an effective manner.

A. FUTURE WORK

Future research can enhance phishing detection strength by augmenting the dataset with additional real-world phishing samples to improve generalizability. DL models, such as CNNs or transformer models, can facilitate automated feature extraction for improved prediction. Furthermore, mechanisms for real-time adaptability will enable the dynamic updating of the model to mitigate evolving threats. In addition, enhancing interpretability and explainability will make the model more transparent, facilitating cybersecurity analysts to better comprehend and analyze threats. Future work will explore deep feature extraction using CNNs and Transformers. Additionally, adversarial testing will be conducted to simulate evasion attacks and test model resilience.

COMPETING INTERESTS

The authors declare that they have no competing interests.

AUTHORS CONTRIBUTIONS

- Saba Naseeb: Conceptualization, data curation, formal analysis, writing—original draft
- Shabana Ramzan: Conceptualization, Methodology, investigation, software, supervision.
- Ali Raza: Conceptualization, Methodology, Writing—original draft, validation.

- Muhammad Shadab Alam Hashmi: Visualization, validation, investigation.
- Yeonghyeon Gu and Muhammad Syafrudin: Conceptualization, Methodology, Supervision, Funding acquisition, Writing—review & editing.
- Norma Latif Fitriyani: Conceptualization, Methodology, Visualization, Writing—review & editing.

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